

Integers, Primes and Roots — Paper A

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is small enough to handle on paper.
- A prime has exactly two factors. 1 is not prime.
- For HCF take the primes in both lists at the lower power; for LCM take every prime at the higher power. As a check, $\text{HCF} \times \text{LCM}$ equals the product of the two numbers.
- Where a question says “in index form”, write $2^3 \times 3^2$, not $2 \times 2 \times 2 \times 3 \times 3$.
- The mark for each question is shown in square brackets on the right.

SECTION 1 – FACTORS, MULTIPLES AND PRIMES

1. Complete the table. Each empty box is worth one mark. [6]

Number	All of its factors	How many factors
18		
28		
36		

Work in pairs from 1 upwards and stop when the pair meets itself. One of these three rows has an odd number of factors — that is not an accident.

2. Write down:

(a) the first four multiples of 7 that are greater than 20 [1]

(b) all the prime numbers between 20 and 30 [1]

3. Write each number as a product of its prime factors, in index form.

(a) 60 [1]

(b) 84

[1]

(c) 200

[1]

SECTION 2 – HCF, LCM, NEGATIVES AND ROOTS

4. Find the HCF and the LCM of each pair. Show the prime factorisations you used.

(a) 12 and 18

[2]

(b) 20 and 30

[2]

5. Work out:

(a) -6×7

[1]

(b) $-48 \div -6$

[1]

(c) $-5 \times -3 \times -2$

[1]

(d) $(-4)^2$

[1]

6. Work out:

(a) $\sqrt{144}$ [1]

(b) the cube root of 64 [1]

(c) 11^2 [1]

SECTION 3 – USING IT

7. Two bells ring at a school. One rings every 8 minutes and the other every 12 minutes. They both ring at 9:00 am. At what time do they next ring together? [3]

Two things repeating and you want them to coincide.

8. Priya has 24 pencils and 36 erasers. She wants to make identical gift packs, using every pencil and every eraser, with as many packs as possible. [3]
How many packs can she make, and what does each pack contain?

9. At midnight the temperature is -4°C . It then falls by 3°C every hour for the next 5 hours. What is the temperature at 5 am? [2]

End of paper. Check every answer has a unit or a form the question actually asked for.